



FINAL PRESENTATION

**Large Multimodal Model Clinical Diagnosis Support
& Reasoning System for Tropical Diseases**

LECTURE:

Dr. Tran Duc Khanh

PRESENTED BY:

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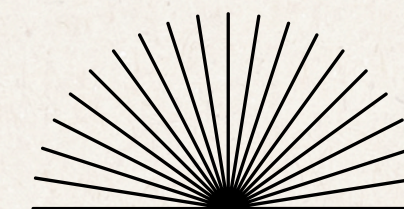
Nguyen Thanh Tu - 10422080

Luong Tieu Cuong - 10422089

Nguyen Thanh Son - 10422072

Nguyen Thien Nguyen - 10422059

Nguyen Hoang Yen Ngoc - 10422056



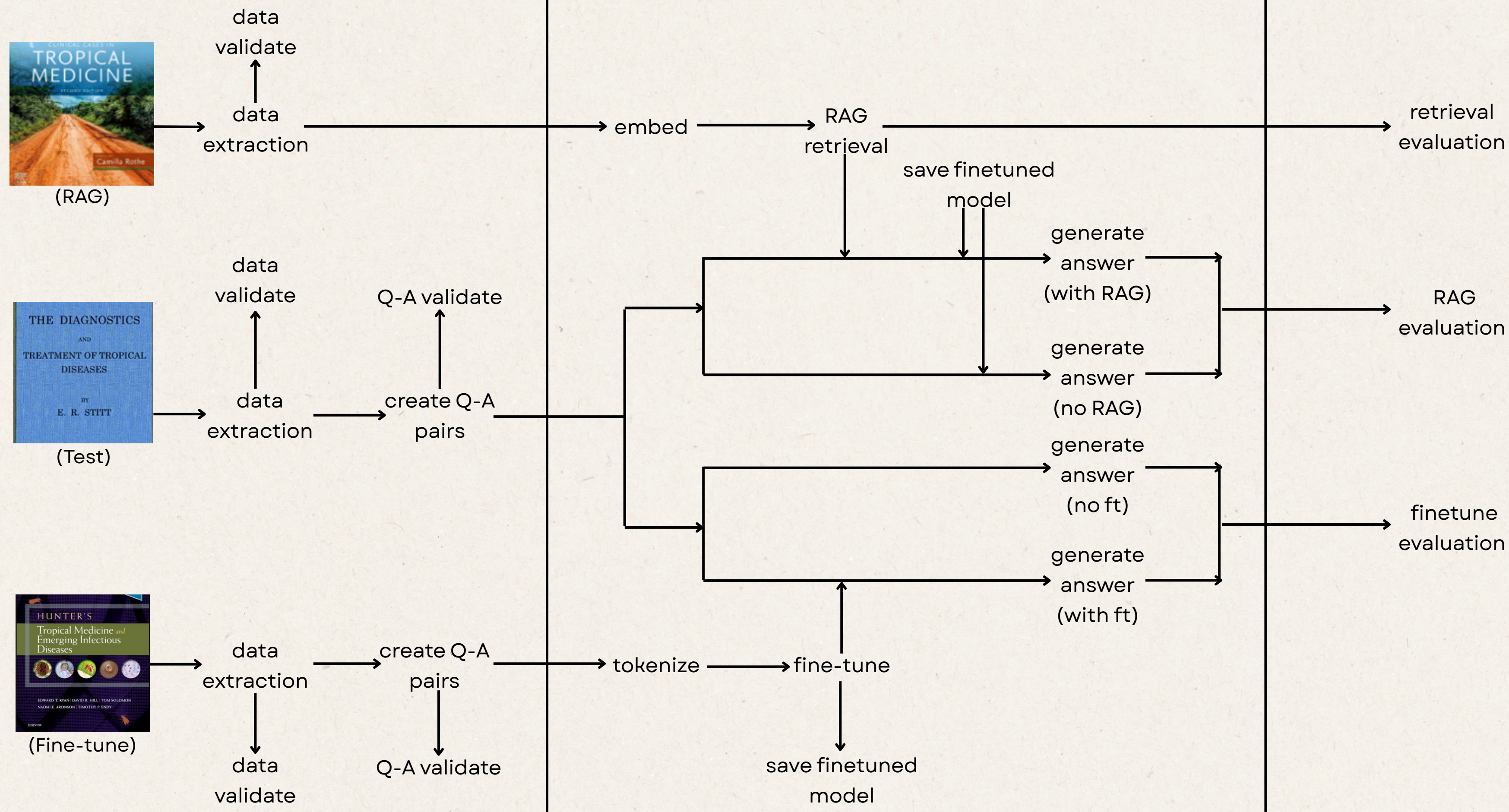
Contributions

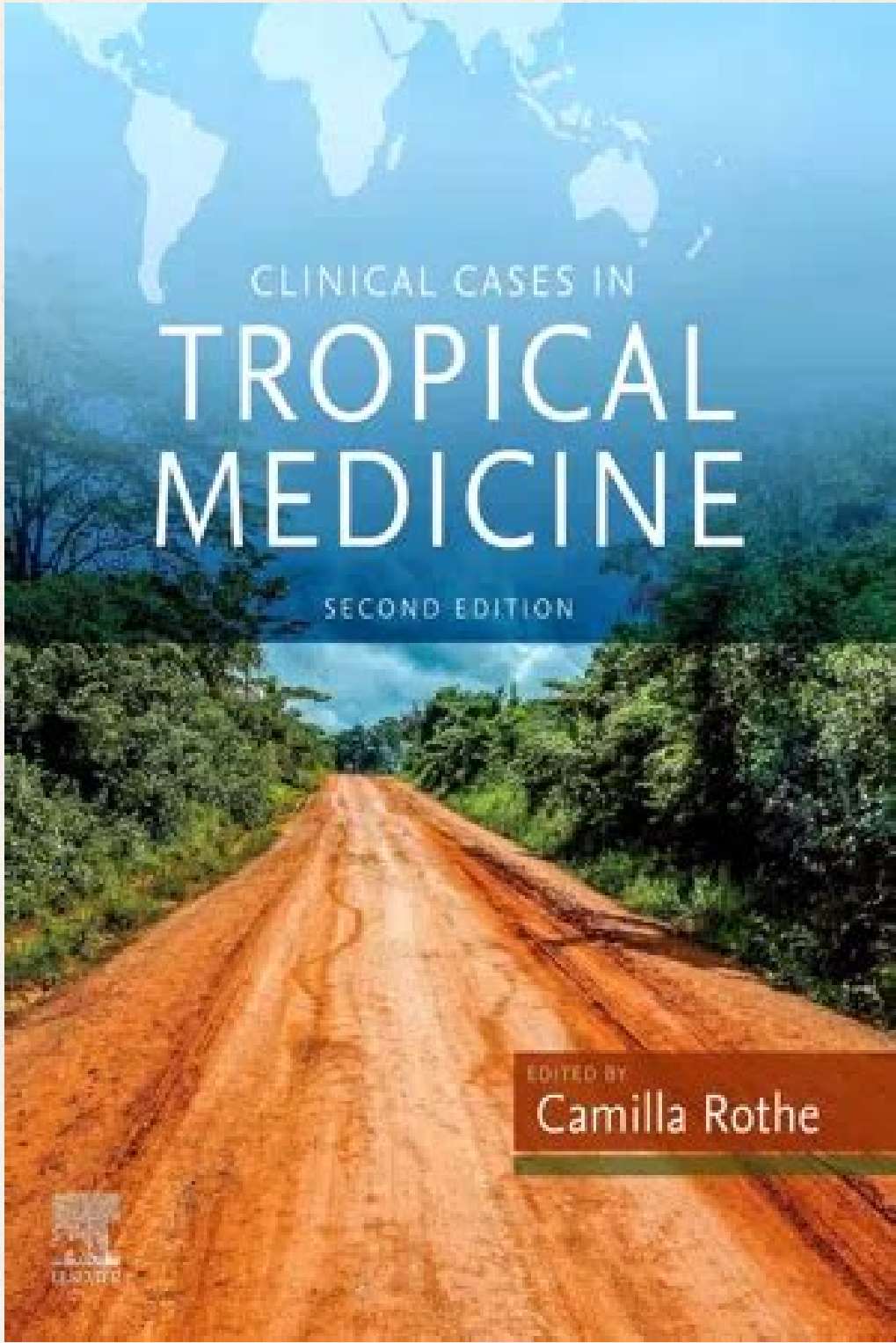
Members	Tasks	Demonstration
Nguyễn Thanh Sơn	Data Extracted Validation 3 Medical Textbooks	Baseline LLaVA-NeXT with prompt inference
Nguyễn Thiên Nguyên	RAG Implement, Baseline LLaVA-NeXT & LLaVA-NeXT with RAG answer generation	LLaVA-NeXT with RAG inference
Nguyễn Hoàng Yến Ngọc	The Diagnostic & Treatment of Tropical Diseases Book Extraction, Test-QA generation	Baseline Medgemma with prompt inference
Võ Tấn Sang	Data Scientist's 93 clinical cases extracted, Hunter's Book extracted	MedGemma with RAG inference
Lương Tiểu Cương	Validate Test-QA/tokenize-QA, finetune LLaVA-NeXT	Finetuned LLaVA-NeXT inference
Nguyễn Thanh Tú	Evaluate RAG/Finetune/RAG+Finetune, RAG Implement, tokenize-QA generation, Final RAG+Finetune output generation	Finetuned LLaVA-NeXT with RAG inference

DATA
PREPARATION

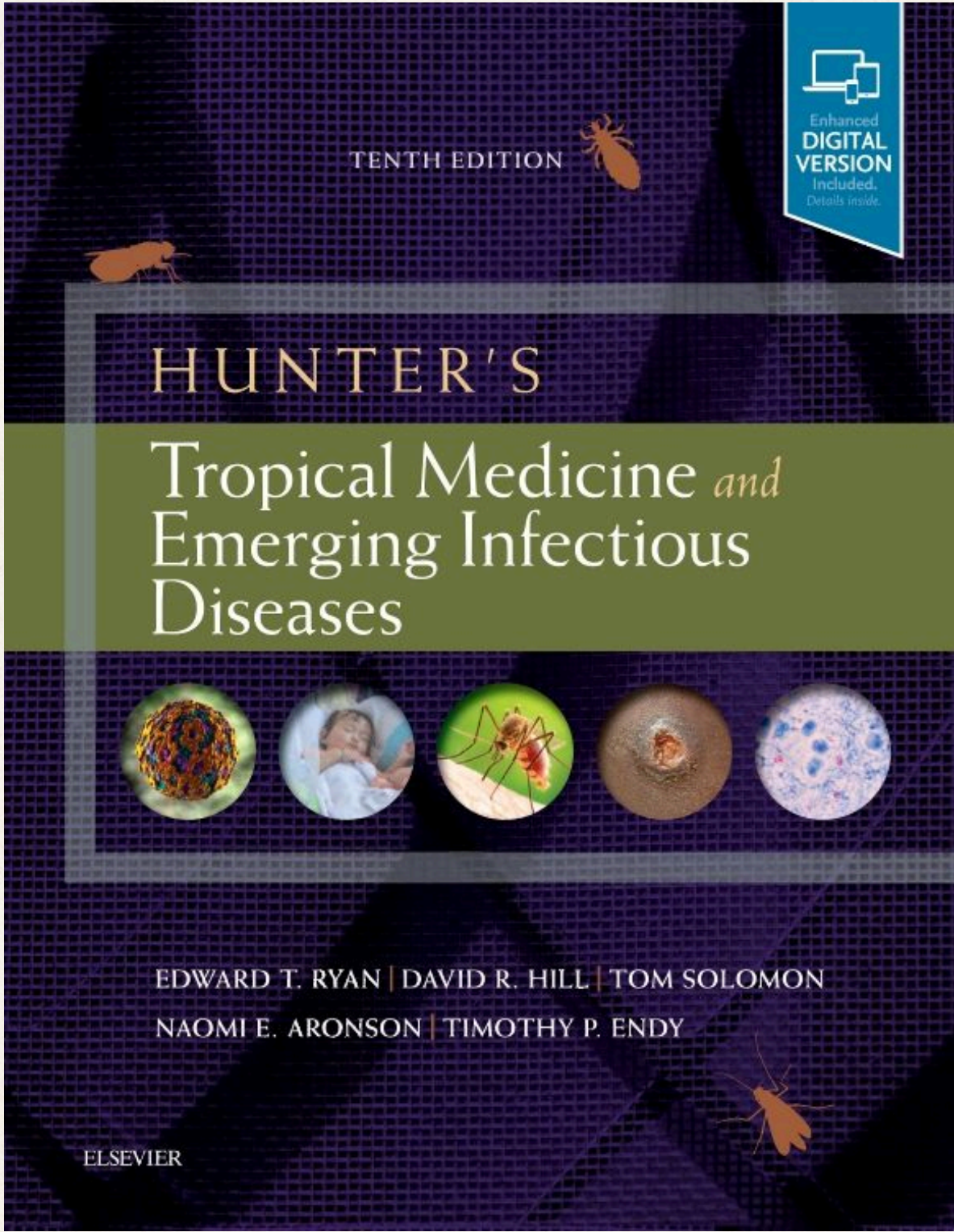
MODEL TRAINING/
RAG/FINE-TUNE

EVALUATION

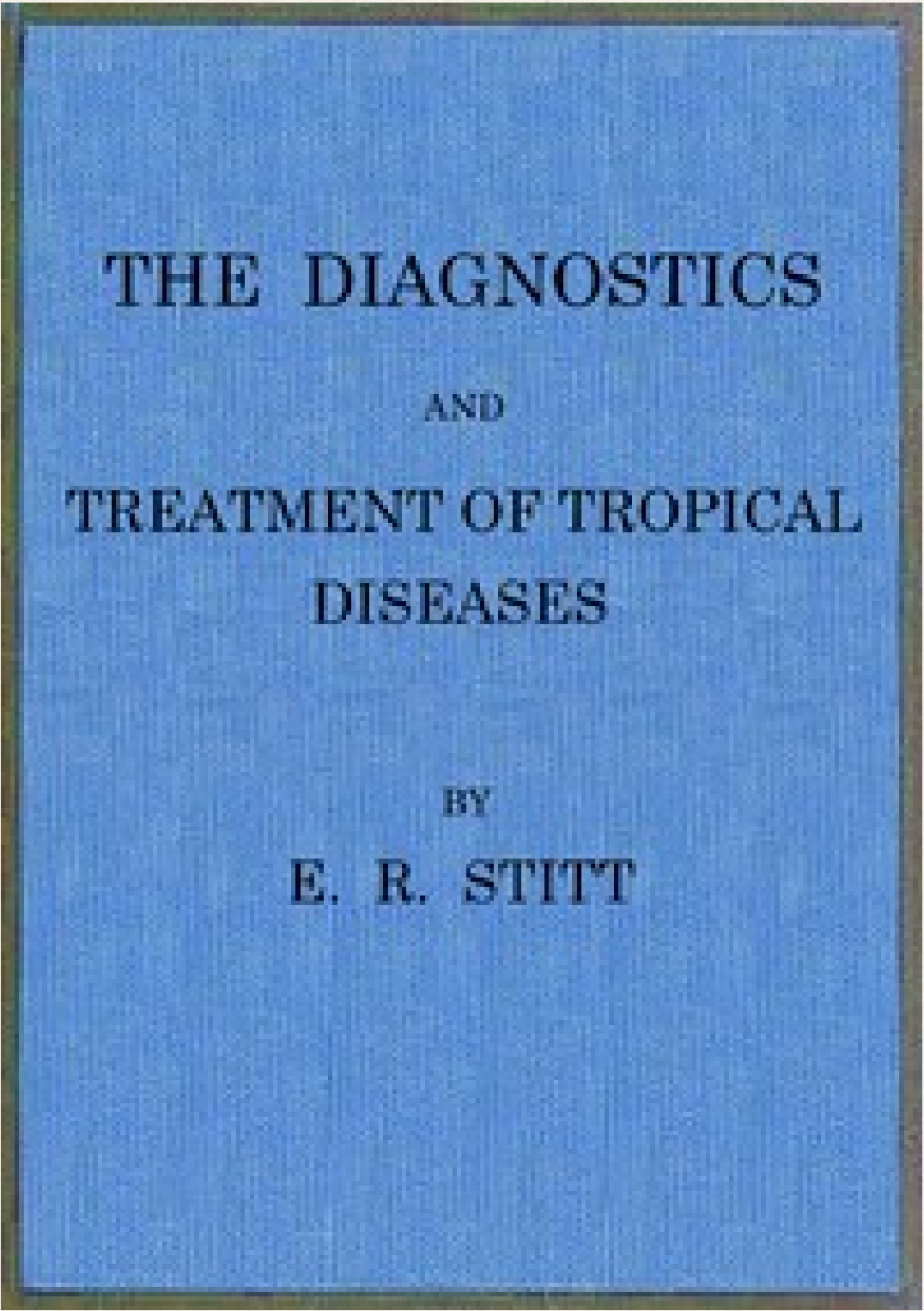




RAG



FINE-TUNE



TEST

DATA CLEANING & TRANSFORMATION

93 clinical cases

```
Project > demo > 93 cases new > extracted_json > {} 37---A-29-Year-Old-Woman-from-Malawi-With-Confusi_2022_Clinical-Cases-in-Tro.js
{
  "patient_information": "A 29-year-old woman from Malawi with an uneventful past medical history. Her H",
  "chief_complaint": "The patient presented with confusion, restlessness, and irritability for the past",
  "history_of_present_illness": "A 29-year-old woman was brought to the hospital by relatives for confus",
  "exposure_and_epidemiology": "The patient lives in a grass-thatched mud-hut in a poor, rural area of M",
  "vitals": "Glasgow Coma Scale was 14/15 (indicating confusion), remaining vital signs were normal, and",
  "physical_exam": "The patient was slim but not wasted. Glasgow Coma Scale 14/15 (confusion). There was",
  "labs_and_diagnostics": "Initial diagnostic workup included an HIV test, which came back negative. A f",
  "differential_diagnosis": "The primary suspected diagnosis, based on the triad of neuropsychiatric cha",
  "management_and_clinical_course": "The patient underwent diagnostic testing, including an HIV test whi",
  "final_diagnosis": "The final diagnosis was pellagra, a nutritional disorder caused by vitamin B3 defi",
  "disease_name_short": "Pellagra",
  "tables": [],
  "images": [
    {
      "file_name": "image1.jpeg",
      "caption": "Hyperpigmented skin rash on sun-exposed skin."
    },
    {
      "file_name": "image2.jpeg",
      "caption": "The skin changes involve both hands and feet. The skin appears dry and scaly."
    }
  ]
}
```

```
s > votan > Downloads > Hunter's > book_extracted_json > {} 33_Viral_Respiratory_Infections.json > ...
{
  "chapter_title": "Viral Respiratory Infections",
  "disease_name": "Viral Respiratory Infections",
  "section_summaries": {
    "epidemiology": "Acute respiratory illnesses (ARIs) are the most frequent il",
    "pathogenesis": "Acute RTIs typically begin in the upper respiratory tract (",
    "clinical_manifestations": "Viral respiratory infections range from common c",
    "diagnosis": "Clinical history and examination offer clues but are non-speci",
    "treatment": "Treatment for mild respiratory infections is often self-limiti",
    "prevention": "Prevention includes frequent handwashing and covering the mou",
  },
  "tables": [],
  "figures": [],
  "references": []
}
```

Hunter’s 67 cases

DATA SYNTHETIC Q-A GENERATION

30 textual QA pairs.

```
Users > HP > Downloads > test Dataset > test Dataset > {} testQA(official)_multi.json > ...
{
  "question": "A 35-year-old male, a European expatriate working for an international aid organization, presents with a two-week history of intermittent fevers, with temperatures typically normal in the morning but spiking to 102.5°F (39.2°C) in the evenings. He also reports experiencing severe headaches, insomnia, and a persistent rapid pulse of 105 beats per minute even at rest. On examination, he is found to have enlarged posterior cervical lymph nodes, described as discrete and non-tender. He notes developing some itchy papules on his trunk and some mild swelling around his eyes. He denies any recent travel outside of his assignment in Kwango Province, Democratic Republic of Congo. What is the diagnosis of this patient?",
  "answer": "West African Trypanosomiasis (Gambiense form) in the early stage",
  "disease_specific_focus": "African Trypanosomiasis",
  "clinical_rationale": "The patient's demographics, including being a European expatriate working in Kwango Province, Democratic Republic of Congo, place him in a region highly endemic for African Trypanosomiasis, a risk factor amplified by his occupational exposure in this area.\n\nThe presenting symptoms strongly suggest the initial stage of African Trypanosomiasis, often referred to as 'trypanosome fever.' The characteristic irregular fever with low morning temperatures and high evening rises (up to 103°F or higher) is present. The rapid, low-tension pulse rate (90-120 bpm) at all times is also a classic finding. Neurological symptoms such as headache and insomnia, along with difficulty concentrating, are early manifestations. The prominent glandular enlargement, specifically the posterior cervical lymph nodes (Winterbottom's sign), is a hallmark of the early stage. The description of itchy papules and localized oedematous swellings, particularly the erythematous rings seen in Europeans, further supports this diagnosis. The incubation period of about two to three weeks aligns with the patient's timeline.\n\nWhile some symptoms like fever and headache could overlap with other febrile illnesses such as malaria or dengue fever, the constellation of findings, particularly the characteristic fever pattern, rapid pulse, posterior cervical lymphadenopathy, and specific skin manifestations in an expatriate from an endemic region, makes African Trypanosomiasis the most probable diagnosis. Other differential diagnoses like syphilis can cause glandular enlargement, but the systemic febrile illness and other specific signs are not consistent. Malaria typically presents with more consistent fever patterns and splenomegaly is more common than posterior cervical lymphadenopathy.\n\nFinal Diagnosis: African Trypanosomiasis",
  "source_file": "CHAPTER_III_TheTrypanosomiases.json",
  "image_path": "Hunters/Book/extracted_json/extracted_images/1000.jpg"
```

SFT Data
~400 samples

```
"source_file": "131_Echinococcosis_Hydatid_Disease.json",
"think": "Key points: This case highlights the importance of considering neurocysticercosis in patients presenting with seizures, espec",
"question": "A 35-year-old cis woman from a rural village in Peru presents to the clinic with a recent onset of focal seizures with sec",
"diagnostic_reasoning": "The patient's presentation with new-onset seizures, particularly focal seizures with secondary generalization,",
"answer_diagnosis": "Neurocysticercosis"
```

DATA EXTRACTED
VALIDATION

Metric (weight)	How we score it	Why it matters in clinical books
Key phrases in summaries (80%)	≥4 literal/near-literal sentences must appear in original PDF	Catches hallucinated WHO stats, wrong sputum colours, fake drug doses
Tables (10%)	Full WHO / parasite tables present → 100 pts None → 0 pts	Tables are the most-used part of every chapter
Figures + References (10%)	Figures listed + any refs → 100 pts Empty → 0 pts	Empty = model skipped half the page
Overall score (out of 100%)	Weighted average → 5 grades	≥ 80% → safe to publish

We replaced “sounds good” NLP metrics with clinically safe, hallucination-proof scoring.

Average Result: 82.1%
=> clinically accurate and complete textually excellent
READY TO USE!!!!

DATA Q-A
VALIDATION

Refernce:

Wu, K., Wu, E., Thapa, R., Wei, K., Zhang, A., Suresh, A., Tao, J. J., Sun, M. W., Lozano, A., & Zou, J. (2025). MedCaseReasoning: Evaluating and learning diagnostic reasoning from clinical case reports. arXiv. <https://arxiv.org/abs/2505.11733>

Prompt 3: Detecting Hallucinations and Inconsistencies in Generated Case Prompts

You are the Editor.
Your job is to audit a draft teaching case that was produced from a published case report. You must confirm strict compliance with all instructions, detect hallucinations, and ensure pedagogic quality.

YOUR INPUTS

1. The original draft you must audit appears between <generated_case>...</generated_case>.

2. The source article appears between <case_report>...</case_report>.

Here is the original guidelines the draft was produced in accordance with:
{convert_case_report_prompt}

CHECKLIST — FAIL ANY ITEM → RAISE A FLAG

A. Source Fidelity

☐ Every fact in each section is traceable to the source article.

☐ No invented details or embellishments.

B. Case Presentation Quality

☐ All facts from the case prompt are present in the source article.

☐ Contains only information known *before* the clinicians formed a differential.

☐ Does **not** reveal the final diagnosis (there should be room for at least some inference).

☐ Provides *sufficient* data (HPI, vitals, exam ± initial tests) for clinicians to formulate a reasonable differential and get the correct final diagnosis.

C. Diagnostic Reasoning Section

☐ Each numbered entry starts with a summary of the reasoning **plus** a direct quote from the article.

☐ Quotes are verbatim or use ellipses (...) without changing meaning.

☐ Paraphrased quotes are okay, as long as they retain the original meaning.

☐ Rationales reference **only information that already appears in** <case_prompt> (not based on new findings, confirmatory tests, or data withheld from students).

D. Final Diagnosis Tag

☐ Final diagnosis is *reasonably* deducible from the case-presentation facts.
— i.e., the final diagnosis should not depend entirely on some test, imaging, or lab result not given in the case presentation.

E. No Hallucinations Anywhere

☐ Every datum, quote, or diagnosis is found in the case report.

Prompt 2: Grading the Quality of Case Reports

You are an expert medical educator tasked with evaluating case reports for their diagnostic-reasoning value. The goal is to find case reports that have a similar style to diagnostic teaching cases from medical textbooks.

CASE REPORT EVALUATION RUBRIC

>> HOW TO USE

1. Read the entire case once without scoring.

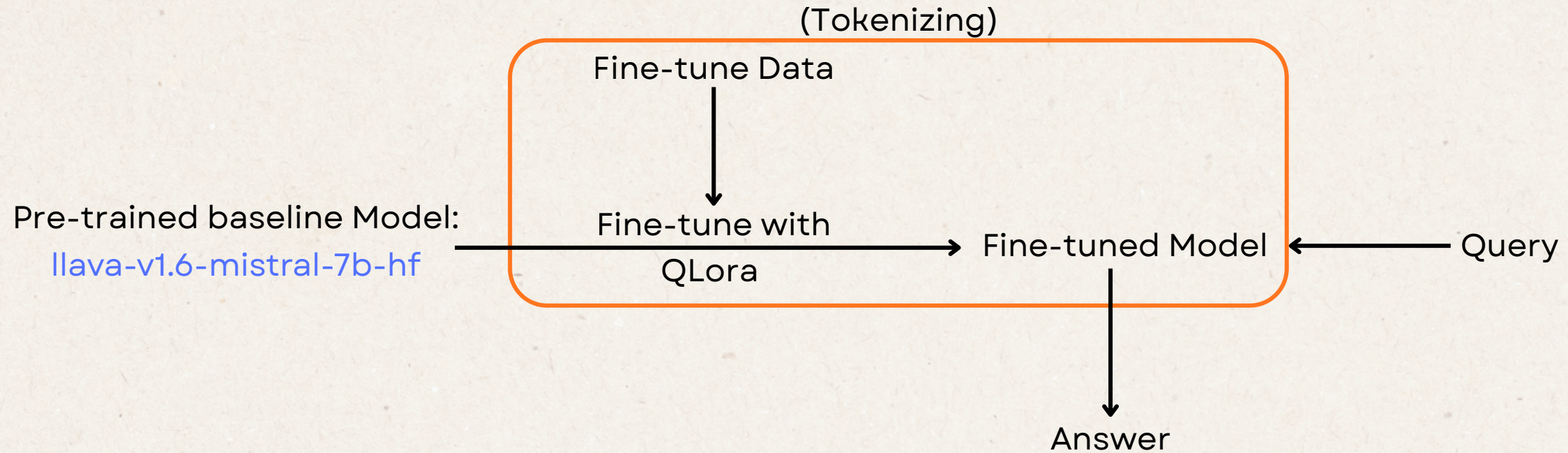
2. Re-read, taking notes.

3. Inside <think>...</think>, write the reasoning that leads you to each score.

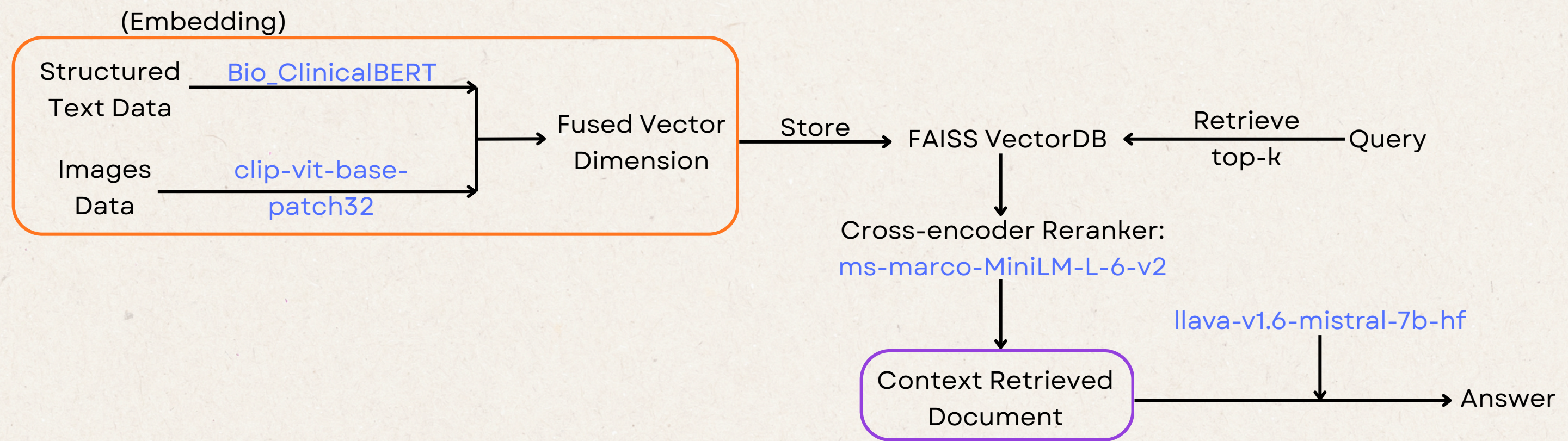
4. Output only the five XML tags shown after the rubric—nothing else.

+-----+
| 1. THOROUGHNESS OF CASE PRESENTATION (1{5 points}) |
| Look for: HPI, past history, meds, allergies, vitals, |
| focused exam, labs, imaging, hospital course, outcome. |
| 1 { Seriously deficient (identifiers only; no vitals) |
| 2 { Major gaps (HPI + vitals OR exam, not both). |
| 3 { Adequate (present but sketchy details). |
| 4 { Very good (complete data, clear timeline). |
| 5 { Exemplary (serial data & course, high quality). |
+-----+
| 2. EXPLICIT DIFFERENTIAL DIAGNOSIS (Yes / No) |
| >=2 plausible alternatives? If yes → "Yes"; else → "No". |
+-----+
| 3. DEPENDENCE ON INTEGRATIVE CLINICAL REASONING (1{5}) |
| Measures need to combine >=2 data points (hx, labs, etc) |
| 1 { Trivial: lone clue gives answer. |
| 2 { Minimal: one dominant clue. |
| 3 { Moderate: must merge TWO findings. |
| 4 { High: THREE+ clues; requires synthesis. |
| 5 { Outstanding: stepwise, complex reasoning. |
+-----+
| 4. TRANSPARENCY OF DIAGNOSTIC REASONING PROCESS (1{5}) |
| 1 { None (no rationale). |
| 2 { Superficial (lists w/o "why"). |
| 3 { Adequate (brief pivots). |
| 4 { Detailed (stepwise, probabilities). |
| 5 { Model (structured, addresses pitfalls). |
+-----+
| 5. STATED FINAL DIAGNOSIS (Yes / No) |
| Is diagnosis clearly named? Yes → "Yes"; else → "No". |
+-----+

**FINETUNE
PROCESSING**



**RAG
PROCESSING**



EVALUATION
RESULTS

Result on 23 test cases including medical cases and images

Method	Models	Expected Output	LLM-As-Judge (Gemini-2.5-flash)			Automatic Metrics		
			Diagnosis Correctness	Reasoning Recall	Faithfulness	Precision	Recall	MAP
Baseline	LLaVA-1.6-7B (prompting)	Answer + Reasoning	0.522	0.529	NA	NA	NA	NA
	MedGemma-4B (prompting)	Answer + Reasoning	0.522	0.626	NA	NA	NA	NA
Finetune	LLaVA-1.6-7B	Answer + Reasoning	<u>0.652</u>	<u>0.675</u>	NA	NA	NA	NA
RAG	LLaVA-1.6-7B	Answer + Reasoning	0.174	0.509	0.705	0.574	0.87	0.636
	MedGemma-4B	Answer + Reasoning	0.478	0.492	0.458	0.6	0.826	0.673
Finetune + RAG	LLaVA-1.6-7B	Answer + Reasoning	0.522	0.602	0.372	0.6	0.826	0.673

INTERPRETATION

- Fine-tuning provides the largest performance gains, improving both diagnosis accuracy and reasoning recall.
- RAG mainly improves evidence grounding, enabling measurable faithfulness when external context is available.
- Fine-tuned + RAG shows no consistent gains over fine-tuning, suggesting reliance on internalized knowledge.
- Faithfulness is NA for non-RAG models, as no external evidence is provided.

LIMITATIONS

- Low faithfulness in Fine-tuned + RAG indicates weak alignment between retrieved cases and model reasoning.
- Retrieval corpus quality limits effectiveness, despite strong retrieval metrics.
- Small multimodal test set (23 cases) reduces result stability.

Thank you for listening